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SG-GAN for Electric Vehicle Road Networks and Q-Learning in Energy Efficient Routing

Suborno Kumar Das, Santi Prasad Maity, and Chinmay Chakraborty, *Senior Member, IEEE*

***Abstract*— This paper suggests a novel routing scheme for multiple electric vehicles (EVs) on a mesh-type road network, which is generated using a Spatial Graph-Generative Adversarial Network (SG-GAN). The scheme focuses on optimizing both the routing of EVs and their association with the respective charging stations(CSs) to reduce energy consumption. The challenge lies in effectively managing factors such as travel cost, the state of charge (SoC) of the battery, and congestion in the road network. To address this problem, we propose a Reinforcement Learning (RL) based approach that minimizes energy consumption and the overall travel time for EVs to reach their destinations. A Q-learning algorithm for RL is employed to solve the nonlinear constrained optimization problems. Through simulation experiments on a road network evolved through SG-GAN, which generates realistic and complex network topologies, we address the allocation of CSs and EVs optimal routing. The results demonstrate the effectiveness of our approach compared to the existing methods, achieving a 2.68% reduction in energy consumption along with a similarity measure, on an average Kullback–Leibler (KL) distance (divergence) 0.54 in the SG-GAN generated road network from the reference real one.**

***Index Terms*— Spatial graph-GAN, Kullback-Leibler divergence, electric vehicle routing, charging station allocation, Q-learning, energy consumption, bipartite graph.**

I. Introduction

THE transition to electric vehicles (EVs) marks a promising and sustainable revolution in transportation. Unlike fossil fuel-dependent vehicles, EVs utilize highcapacity batteries and can be charged from renewable sources such as solar or wind, reducing pollutants and greenhouse gases. This shift is driven by environmental concerns, advancements in lithium-ion battery technology resulting in higher energy densities, longer ranges, and faster charging times [1] and supportive government policies alongside evolving consumer preferences. EVs are central to smart city development, yet challenges such as designing lightweight batteries, efficient power electronics, and optimized motor systems persist, all of which influence energy efficient (EE) routing. EE routing algorithms also depend on the availability and strategic placement of Charging Stations (CS). Reducing energy consumption in EVs extends driving range, minimizes charging frequency, lowers overall ownership costs, and decreases environmental impact. This work addresses the EV routing problem by focusing on generating road networks for multiple EVs traveling from arbitrary source points to fixed destinations while accessing CS along the way.

The rest of the paper is organized as follows: Section II presents a literature review. Section III presents a Spatial Graph-Generative Adversarial Network (SG-GAN) based road network Generation with embedded CS, and Section IV presents energy consumption, travel time, and congestion factor calculations. EE-EV Routing and CS allocation discussion are made in Section V. Results and discussion are made in Section VI. Finally conclusion and future work are presented in Section VII.

II. Literature Review on Related Works

Literature review on intelligent transportation and EV infrastructure presents a diverse set of methodologies and algorithms addressing road network generation, EV charging optimization, routing decisions, and EV to CS allocation. For instance, [2] highlights that the increasing number of vehicles intensifies traffic congestion, air pollution, and energy consumption. Recent advances [3] address issues in spatial and topological modeling by combining transformer-based GANs with spatial and topological supervision, along with differentiable image-to-graph extractors, thereby improving metrics such as traffic convenience. Addressing missing data in real-time EV Charging Monitoring (EVCM) systems, [4] introduced the CurriFusFormer framework, that shows result of missing data rates from 30% to 90%, achieving R^2 scores between 0.83 and 0.92. In parallel, optimal charging strategies for Distribution System Operators have been developed using Markov Decision Processes (MDP) and Reinforcement Learning (RL) [5]. An RL approach has also been applied to the dynamic EV routing problem to minimize energy consumption and prevent battery depletion [6]. Other contributions include Actor-Critic-based smart charging algorithms to mitigate EV charging impacts on power grids [7] and Deep Reinforcement Learning (DRL) approaches for the optimal route and CS selection under dynamic traffic conditions and uncertain future requests [8]. DRL-based approaches have minimized total travel time and charging costs [9], while multi-agent RL has been used for optimal energy scheduling

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at CSs, enhancing economic profits and user satisfaction [10]. Complementary strategies include an RL-based EV assignment scheme (RLEVAS) designed to reduce charging costs and grid overload [11], and DRL methodologies for an optimal CS placement in urban areas that maximizes charging plan benefits [12]. Additionally, DRL has been employed for dynamic energy scheduling and routing of multiple EVs to lower energy costs [13], with multi-agent DRL approaches extending to EV charging scheduling in smart grids [14]. The green vehicle routing problem has been examined in [15], while efficient EV route planning using the Bellman-Ford algorithm on an embedded GPU is presented in [16]. Hierarchical and hybrid multi-agent RL approaches have been proposed to optimize EV flexibility services [17], and a bilevel graph RL method was developed for effective EV fleet charging guidance [18]. An end-to-end RL-based routing method for electric logistics vehicles further demonstrates reduced energy consumption in [19]. Recent research extends to motion planning for EVs using RL to optimize both battery consumption and travel time efficiency [20].

A. Scope and Contributions of the Work

EV routing in smart cities faces challenges due to newly emerging dynamic road networks, fluctuating traffic conditions, and limited battery capacities [21], [22]. Motivated by [3], [6], [7], and [9], the present work integrate SG-GAN, Q-learning, and a bipartite graph method with contributions as follows:

- We employ SG-GAN to generate a Generative AI-based dynamic and realistic road network, enabling EVs to navigate unknown and evolving road networks efficiently. With a KL Distance (Divergence) value of 0.54 [23], the generated network resembles closer to the reference real road network.
- We develop a Q-learning algorithm, an RL-based method that optimizes travel time and energy consumption by dynamically learning from real-time traffic conditions using a priority-based optimization framework. This approach reduces energy consumption by 2.68% compared to the method in [22].
- The present work also ensures seamless CS allocation using the concept of bipartite graph mapping, reducing range anxiety and improving EV mobility in smart cities.

III. SG-GAN Based Road Network Generation With Embedded CS

We propose an SG-GAN [2]-based architecture to synthesize realistic road networks from OpenStreetMap (OSM) data, where road layouts, intersections, road types, and connectivity are encoded as a graph.

$$G_r(V_e, E_d), \quad V_e = \{v_1, v_2, \ldots, v_n\}, \quad E_d = \{e_{ij}\},$$

where edges are weighted by attributes such as distance, travel time, and congestion. Representation includes an adjacency matrix, an edge weight matrix, and a connectivity probability distribution.

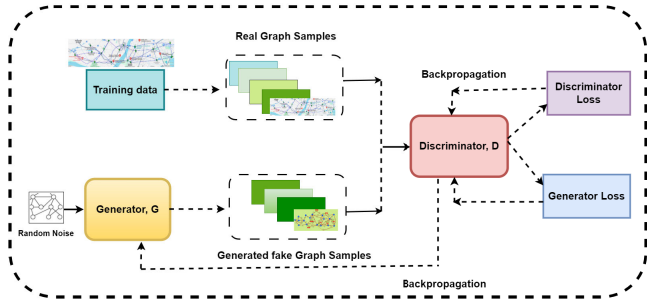

Fig. 1. Proposed SG-GAN architecture.

A. Model Structure, Training and Connectivity Modeling

The SG-GAN framework consists of a generator G and a discriminator D that engage in adversarial training. The generator approximates the true conditional connectivity distribution $p_{\text{true}}(v \mid v_c)$ for a given vertex v_c, while the discriminator distinguishes real samples from those synthesized by G. Their training objective follows the minimax formulation [2]:

$$\min_{\theta_G} \max_{\theta_D} \mathcal{L}(G, D) = \sum_c \mathbb{E}_{v \sim p_{\text{true}}(\cdot \mid v_c)} \left[\log D(v, v_c; \theta_D)\right] + \mathbb{E}_{v \sim G(\cdot \mid v_c; \theta_G)} \left[\log\big(1 - D(v, v_c; \theta_D)\big)\right]. \tag{1}$$

Weights in both G and D are initialized using Xavier (Glorot) initialization as

$$U \sim X\Big(-\frac{6}{p_{\text{in}} + p_{\text{out}}}, \frac{6}{p_{\text{in}} + p_{\text{out}}}\Big), \quad bias = 0.$$

Pre-training involves training G on a reconstruction task to learn an initial embedding of road network features and training D on a binary classification task (real vs. generated graph samples). This warm-up (upto 10 epochs) precedes full adversarial training. Optimization employs the Adam optimizer with $\alpha = 0.0002$ (starting at $\alpha = 0.0001$ for the first 50 iterations), $\sigma_1 = 0.5$, $\sigma_2 = 0.999$, along with gradient clipping (max norm 5.0) and batch normalization. To capture connectivity, a graph softmax function [2] is introduced:

$$G(v \mid v_c) = \prod_{j=1}^{m} p_c(v_{r_j} \mid v_{r_{j-1}}) \cdot p_c(v_{r_{m-1}} \mid v_{r_m}),$$

where each vertex v_n has neighbors $\mathcal{M}(v_n)$ sampled from the conditional distribution $\tilde{T}(v \mid v_n)$. The SG-GAN frame work dynamically adapts road networks based on real-time conditions such as traffic fluctuations, roadwork, or obstacles. This makes it a powerful tool for adaptive urban planning and smart city development. Our proposed SG-GAN architecture is shown in Fig.1 and the SG-GAN-based road network generation algorithm is shown in Algorithm 1.

B. KLD-Based Assessment of Generated Networks

To quantitatively compare the generated and real road networks, we employ the Kullback-Leibler (KL) divergence [23]. For each edge (u, v), we compute the distance-to-time ratio R_{uv} as:

$$R_{uv} = \frac{\text{distance}_{uv}}{\text{time}_{uv}}. \tag{2}$$

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The ratio is normalized to [0, 1] via min-max normalization as r'_{uv}:

$$R'_{uv} = \frac{R_{uv} - \min(R)}{\max(R) - \min(R)}. \tag{3}$$

Algorithm 1 SG-GAN for Road Network Generation

Require: Dimension of embedding k, size of generated samples s, size of discriminated samples t

Ensure: Generator $G(v|v_c;\theta_G)$, Discriminator $D(v, v_c;\theta_D)$

```
Initialize and pre-train G(v|v_c; θ_G) and D(v, v_c; θ_D)
Construct Breadth-First Search (BFS) tree T_c for all v_c ∈ V
while SG-GAN has not converged do
    for each G-step do
        G(v|v_c; θ_G) generates s vertices for each vertex v_c
        Update θ_G by minimizing the generator loss
    end for
    for each D-step do
        Sample t positive vertices from the ground truth road network
        Sample t negative vertices from G(v|v_c; θ_G) for each vertex v_c
        Update θ_D by minimizing the discriminator loss
    end for
end whilereturn G(v|v_c; θ_G) and D(v, v_c; θ_D)
```

Let N be the number of intervals used for partitioning the normalized values. The total count of samples M is defined as

$$M = \sum_{j=1}^{N} n_j, \tag{4}$$

where n_j represents the number of occurrences in the j-th interval. Probability of the j-th interval is defined as

$$p_j = \frac{n_j}{M}. \tag{5}$$

The KL divergence is then given by

$$D_{KL}(p \parallel q) = \sum_{j=1}^{N} P(j) \log \frac{p(j)}{q(j)}.$$

where p is the probability distribution of a feature in a real road network and q is the probability distribution of a feature in a SG-GAN-generated road network.

IV. Calculation of Energy Consumption, Travel Time and Congestion Factor

In the present routing problem, energy consumption (E_c) is defined in the Eq.6 for the i_{th} EV [24].

$$E_{\text{ci}} = \frac{1}{\eta}\left(\frac{1}{2}\rho C_d A v^2 + C_r m g v + m g d \sin\theta\right) \tag{6}$$

where

- η = Mechanical to electrical energy conversion efficiency
- ρ = Air density (in kg/m^3)
- C_d = Drag coefficient (aerodynamic resistance factor)
- A = Frontal area of the vehicle (in m^2)
- V = Velocity of the vehicle (in m/s)
- C_r = Rolling resistance coefficient.
- m = Mass of the vehicle (in kg)
- g = Acceleration due to gravity (approximately 9.81 m/s^2)
- d = Distance
- θ = Angle of incline ($\sin(\theta)$ accounts surface slope)

Travel time (T_c) defined in the Eq.7 for the i_{th} EV as

$$T_c = \frac{L}{\lambda} + \frac{d}{V} + T_{\text{charge}} \tag{7}$$

where queueing delay (W) given by Little's Law [25]:

$$W = \frac{L}{\lambda} \tag{8}$$

and charging time at each CS (T^s_{charge}) given by:

$$T^s_{\text{charge}} = \frac{B_{\max} - B_{\text{current}}}{\eta_s} \quad \forall s \in \text{CS}$$

- T^s_{charge} is the charging time at CS(in hours),
- $B_{\max}$ is the maximum SOC of the EV (in kWh),
- B_{current} is the current SOC upon arrival at the station (in kWh),
- η_s is the charging rate at station s (in kW)

Congestion factor for a road segment (i, j) is computed as

$$\delta_{ij} = \frac{q_{ij}}{C_{ij}}, \tag{9}$$

where q_{ij} denotes the traffic flow (i.e., the number of vehicles per unit time on road segment (i, j)) is modeled as a Poisson-distributed random variable between 0 to C_{ij} where C_{ij} represents the maximum number of vehicles that the segment can accommodate per unit time. This ratio is normalized between 0%(minimal traffic flow) and 100% (complete congestion). Our framework measures local congestion, δ_{ij}, which reflects the congestion level on an individual road segment. This local metric supports fine-tuned, immediate routing decisions to avoid short-term bottlenecks. Consequently, a higher δ_{ij} value on a given segment prompts EV agents to dynamically adjust their routes, thereby enabling an adaptive response to both overall and localized traffic disturbances. Global congestion δ calculated using the Eq.10 where w_{ij} represents the weight on the road segment, assumed uniform.

$$\delta = \frac{\sum_{(i,j)} w_{ij}\,\delta_{ij}}{\sum_{(i,j)} w_{ij}} \tag{10}$$

Global congestion aggregates the congestion factors of all road segments across the network to provide an overall assessment of traffic density based on the positions and trajectories of all EV agents.

A. Defining the Objective Function

We consider the overall objective function O for n no. of EVs, formulated as a weighted sum of energy consumption

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and journey completion time. The objective function is given in the Eq.11.

$$O = \min \sum_{i=1}^{n} \left(\beta \frac{E_{c_i}}{E_{\text{norm}}} + (1-\beta) \frac{T_{c_i}}{T_{\text{norm}}} \right) \tag{11}$$

where $\beta \in [0, 1]$ is a dimensionless weighting factor that controls the trade-off between the two $\beta = 1$, energy minimization and $\beta = 0$ travel time minimization). Normalization factors E_{norm} and T_{norm}, typically set to their maximum expected values, ensure that

$$0 \leq \frac{E_{c_i}}{E_{\text{norm}}}, \frac{T_{c_i}}{T_{\text{norm}}} \leq 1,$$

thereby scaling both terms comparably. The optimization is subject to the following constraints:

$$\sum_{i=1}^{N} y_{sj} \leq C_j \quad \forall \text{CS}_j,$$

where $y_{sj} \in \{0, 1\}$ indicates whether i_{th} EV is charged at CS_j, and C_j represents the available charger count at station CS_j. When this count reaches the capacity C_j, additional EVs incur a queuing delay W_i.

V. Energy Efficient EV Routing and CS Allocation

We propose a decentralized multi-agent Q-learning framework for routing EVs in a dynamic road network. Each EV operates as an independent agent, making routing and charging decisions based on its local observations solely while contributing to a global context by updating real-time congestion and CS mappings.

A. Algorithmic Implementation

1) Initialization: Each EV agent initializes its Q-table from arbitrary starting locations with zeros, selects an initial battery SOC from a continuous uniform distribution, and is assigned a fixed destination. A global set S records all visited nodes along the optimal paths, while a bipartite mapping Bip dynamically associates high-traffic zones with nearby CSs. A global congestion parameter δ captures the evolving density of EVs across the network.

2) State, Action, and Reward: Let the state s be the EV's current position in the network and the action a is movement to a neighboring node or accessing a CS. The reward for the ith EV is defined as:

$$Reward_i = R_{\text{base}} - \left[\beta \frac{E_{ci}}{E_{norm}} + (1-\beta) \frac{T_{ci}}{T_{norm}} \right] K_{\text{obj}} - \frac{\delta_{ij}}{\delta_{norm_{ij}}} K_{\text{congestion}} + I_{\text{dest}} K_{\text{bonus}}, \tag{12}$$

where

- R_{base}: Base reward.
- K_{obj}: Scaling factor for the weighted cost.
- $K_{\text{congestion}}$: Scaling factor for the congestion penalty.
- I_{dest}: Indicator (1 if destination is reached, 0 otherwise).
- K_{bonus}: Bonus awarded upon reaching the destination.
- δ_{norm}: Maximum expected congestion for EV.

Algorithm 2 Q-Learning for EV Routing and CS Allocation

Input: SG-GAN generated Weighted Graph $G(V, E)$ from Algorithm 1, α, γ, ϵ, θ

Output: Set of optimal routes, cumulative energy consumption, total travel time

Initialization:
for each EV agent $i \in \{1, \ldots, l\}$ **do**
  Initialize Q-table: $Q_i(s, a) \leftarrow 0$ for s and a
  Set initial energy, SOC, and destination for agent i
end for
Initialize bipartite mapping $Bip \leftarrow \emptyset$
Initialize global visited node set $S \leftarrow \emptyset$
Initialize global congestion δ w.r.t EV densities over G
for all EV agents i in parallel **do**
  while agent i has not reached its destination **do**
    From current state s, select action a using an ϵ-greedy policy with $Q_i(s, a)$, considering current congestion from δ
    Execute action a; observe new state s', energy consumption E_{ci}, travel time T_{ci}, local congestion factor δ_{ij} (from δ), and update battery SOC
    Compute reward function

$$Reward_i = R_{\text{base}} - \left[\beta \frac{E_{ci}}{E_{norm}} + (1-\beta) \frac{T_{ci}}{T_{norm}} \right] K_{\text{obj}} - \frac{\delta_{ij}}{\delta_{norm_{ij}}} K_{\text{congestion}} + I_{\text{dest}} K_{\text{bonus}}$$

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha \left[R_i + \gamma \max_{a'} Q_i(s', a') - Q_i(s, a) \right]$$

    Add the newly visited node s' to S
    Extract S' from S, a subset of high-traffic nodes
    For each node $v \in S'$, dynamically map v to the nearest available CS using bipartite matching; update B
    if s' **and** EV battery SOC $< T_h$offers multiple feasible CS options $\{CS_j\}$ **then**
      for each available CS CS_j **do**
        Compute combined metric:

$$CS_j = \beta \cdot \frac{\Delta E_j}{E_{\text{norm}}} + (1-\beta) \cdot \frac{W_j}{W_{\text{norm}}}$$

      end for
      Select $CS^* \leftarrow \arg\min_j \{C_j\}$
      Update mapping $B \leftarrow B \cup \{(s', CS^*)\}$
    end if
    Update δ global congestion.
    Set current state $s \leftarrow s'$
  end while
end for
return {Optimal Routes, Cumulative Energy Consumption, Total Travel Time}

3) Learning Process and Q-Value Update: At each epoch, an EV agent observes its state s and selects an action a using

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an ϵ-greedy policy, which balances exploitation of learned Q-values and exploration of new routes while incorporating local congestion measures δ_i (derived from the global δ). After executing a, the agent transitions to a new state s', updating its battery SOC and collecting data on energy consumption E_{c_i}, travel time T_{c_i}, and local congestion δ_i. The Q-values are updated according to:

$$Q_i(s,a) \leftarrow Q_i(s,a) + \alpha \left[R_i + \gamma \max_{a'} Q_i(s',a') - Q_i(s,a) \right], \tag{13}$$

where α is the learning rate and γ is the discount factor. The optimal Q-value function and policy are defined as:

$$Q^*(s,a) = \max_{\pi} Q^{\pi}(s,a), \pi^*(s) = \arg\max_{a} Q^*(s,a). \tag{14}$$

Using ϵ-greedy policy [26], the EV selects the greedy action:

$$a = \arg\max_{a \in \mathcal{A}} Q_t(a), \tag{15}$$

The action selection probability is given by:

$$\pi_t(a) = \begin{cases} 1 - \epsilon + \dfrac{\epsilon}{|\mathcal{A}|}, & \text{if } Q_t(a) = \max_{b \in \mathcal{A}} Q_t(b), \\ \dfrac{\epsilon}{|\mathcal{A}|}, & \text{otherwise.} \end{cases} \tag{16}$$

4) Charging Station Allocation via Bipartite Graph: A bipartite graph is employed to optimize CS allocation. Two disjoint sets are defined: S', representing high-frequency intersections, and CS, consisting of CS locations generated by SG-GAN. Edge weights between S' and CS are computed as:

$$CS_j = \beta \cdot \frac{\Delta E_j}{\Delta E_{\text{norm}}} + (1-\beta) \cdot \frac{W_j}{W_{\text{norm}}}, \tag{17}$$

where ΔE_j is the additional energy required to reach station j, W_j is the expected queuing delay, and ΔE_{norm} and W_{norm} are normalization factors ensuring unitless metrics. When an EV at state s' has multiple CS options and its battery SOC is below a threshold θ, the assignment is determined using the Hungarian algorithm [27]. Our proposed RL-based method for optimal routing and CS placement is shown in Algorithm 2.

VI. Results and Discussion

In our experiment, a mesh-like topology for road networks is generated by SG-GAN with the number of junctions nodes 29, and 7 CS nodes and 50 EVs. Simulations were implemented in Jupyter environment using Python, and the simulations were executed on a personal computer featuring an Apple M1 Silicon Chip and 8 GB of RAM. Numerical values for the parameters are shown in Table I and hyperparameters to run the proposed scheme are shown in Table II.

By modifying the distance and time parameters, two road networks using SG-GAN are shown in Fig.2(a) and (b), a wider and denser network, respectively. Each EV is initialized in the road network with a uniform 10 kWh battery and an initial state-of-charge (SOC) uniformly distributed between 30% and 100%, averaging 65% with a standard deviation of 20.2%. The algorithm computes optimal routes and reroutes EVs to the nearest CS when their SOC falls to a critical threshold, ensuring a fixed energy buffer of 2.5 kWh in our experiment to prevent depletion. For a 10 kWh battery, this corresponds to a 25% SOC threshold. To maintain the same safety margin across varying battery sizes, the threshold is adjusted proportionally as 16.7% for 15 kWh, 12.5% for 20 kWh, and 8.3% for 30 kWh. This prevents early rerouting while ensuring consistent energy reserves before charging.

For the road network shown in Fig.2(a), Table III shows the different optimal paths with CS allocation for 5 EVs without using bipartite graph mapping. These paths are evaluated at different congestion levels (25%, 50%, and 100%), where 100% represents the maximum number of EVs in the network, following a Poisson distribution with a mean arrival rate of 5 EVs. In our observation increase on energy consumption and travel times is found to be higher due to longer paths because there is no optimal mapping of EV to CS (EV randomly mapping to CS). When the bipartite graph is used, under 25% congestion, energy consumptions (kWh) for the EVs are found to be 4.66, 1.25, 4.46, 0.86, and 4.63, respectively. At 50% congestion, these values increase to 5.34, 1.33, 5.13, 0.93, and 5.31 kWh. At 100% congestion, the values

TABLE I
Parameters Related to Energy Calculation

Parameter	Description	Value
ρ	Air density	$1.225\,\text{kg/m}^3$
C_d	Drag coefficient	0.3
A	Frontal area of the vehicle	$2.5\,\text{m}^2$
C_r	Rolling resistance coefficient	0.01
m	Mass of the EV	1500 kg
g	Acceleration due to gravity	$9.81\,\text{m/s}^2$
η	Mechanical to electrical energy conversion efficiency	0.9
θ	Angle of the incline	5°

TABLE II
Hyperparameters of the Proposed Strategy

Parameter	Description	Value
Network Type	Network used for experiments.	Mesh
Initial Nodes	Initial number of nodes in the network.	29
Maximum Episode	Cutoff training episode number.	10000
Learning Rate(α)	Sets the Q-value update step size.	0.1
Exploration	Initial exploration rate in ϵ-greedy.	1
Cutoff Exploration	Cutoff exploration rate in ϵ-greedy.	0.1
Gamma (γ)	Discount factor.	0.75
Beta (β)	Weighting factor	0.8.
Batch size		64
Optimizer(σ)	Adam	0.5 and 0.99
(Th)	EV battery SOC threshold	25%
B_{Max}	EV battery capacity	10 KWh
C_j	Available Charger Count per CS	2

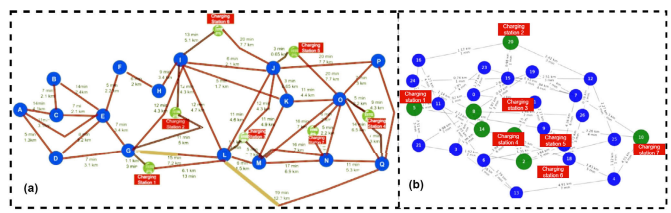

Fig. 2. Two SG-GAN generated road networks with 22 junction nodes (blue color) and 7 CS (green color), Inclined path (yellow color) by changing parameters distance and time; (a) Wider network and (b) Dense network.

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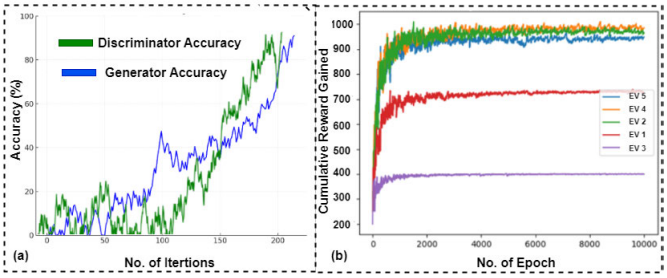

Fig. 3. Learning Curves (a) Accuracy for the Generator and Discriminator, (b) Cumulative Reward gained graph for 5 EVs.

rise to 6.12, 1.42, 5.81, 0.99, and 6.10 kWh, respectively shown in Table IV. The average energy consumption without bipartite mapping is found 18.41 kWh/100 km, compared to 15.64 kWh/100 km with bipartite graph mapping, showing a 15.05% savings in energy consumption. For the road network shown in Fig.2(b), we analyze the resulting changes in energy consumption and travel time for 5 EVs using bipartite mapping, which are presented in Table V. On average 37.6% more energy consumption and 2.26% more travel time needed at 50% congestion for road network in Fig. 2(b) compared to the road network in Fig.2(a) for EVs to reach their destination using the optimal paths beacuse road network in Fig.2(b) has a denser structure, As a result, frequent junctions and routing through many short segments and tight turns force repeated deceleration and re-acceleration. Each cycle incurs additional mechanical work and energy losses compared to steady cruising. Moreover, the dense network leads to higher local congestion, resulting in queues and idling delays, further increasing travel time.

Now the training progression of SG-GAN is depicted in Fig.3(a). During the initial 50 iterations, generator accuracy fluctuates between 45% and 65%, while discriminator accuracy ranges from 40% to 75% showing significant instability. By iteration 100, the generator reaches 82% (Â±3%) and the discriminator stabilizes at 85%. At iteration 150, the generator achieves 89% (Â±2%) and the discriminator 90%. Upon convergence at iteration 200, the generator stabilizes at 91% (Â±1%) and the discriminator at 93%, with a narrow 2% performance gap. Notably, after the 50th epoch, the generator improves at an average rate of 0.23% per iteration compared to the discriminator 0.18%, confirming robust convergence with exceeding 90% accuracy. Similarly Fig.3(b) shows the cumulative reward gained graph for 5 EVs, where reward increases steadily with the number of iterations, and stabilizes after 2000 iterations. Although the reward gained graph looks similar, the cumulative reward values are different. For example, the reward values of EV5, EV4, and EV2 (Group II) are much higher than EV1 and EV3 (Group I), possibly the better optimal possible paths availed by Group I than Group II, and due to the non-use of CS.

Fig.4(a), shows a three-dimensional (3D) plot indicating the congestion factor, mean completion time, and mean energy consumption values, with each point corresponding to three parameters, A (100, 16.66, 404.57), B ((50, 15.64, 333.33), C ((25, 13.83, 280.95). It is evident that an increase in congestion

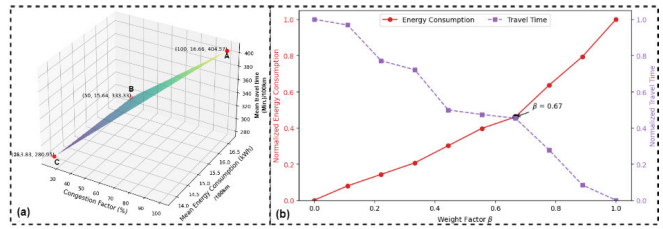

Fig. 4. (a) 3D plot of Congestion, Mean energy and Travel time, (b) Sensitivity analysis on β value.

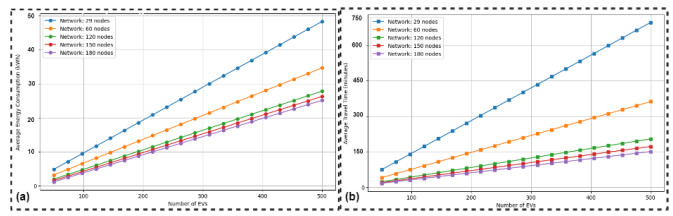

Fig. 5. For the different Network size (a) No.of EV vs Energy consumption (b) No.of EV vs Travel time.

leads to higher energy consumption and takes more time to complete its journey. Point B seems to be a preferable operating point that balances among three factors over the points A and C.

Fig.4(b) presents a sensitivity analysis of the weighting factor β. A low value ($\beta = 0.1$) prioritizes shorter travel times, while a high value ($\beta = 0.9$) emphasizes energy savings. Around $\beta = 0.67$, the two objectives are balanced, but since our focus is on energy efficiency, we select $\beta = 0.8$ to favor reduced energy consumption. We see in the plot that travel time is highly sensitive to β, so that small adjustments can changes in routing or speed, producing a significant time variability. In contrast, energy consumption responds more gradually, as efficiency gains follow smoother physical laws (moderate acceleration, steady speeds) and operate within the battery's fixed capacity. Thus, energy optimization yields smooth transitions, whereas travel time can fluctuate sharply due to traffic dynamics and route changes.

Fig.5(a) and Fig.5(b) show that as the number of EVs on the road increases, both energy consumption and travel time rise due to high congestion, longer queuing delays at CS, and the need for EVs to take alternative routes. However, larger road networks with a larger number of nodes and CS experience comparatively less congestion. These expansive networks can better accommodate the growing demand, resulting in lower energy consumption and reduced travel time.

Table VII compares EV routing methods under comparable urban, battery-aware simulation conditions. Clustering historical driving data [28] achieves an average energy consumption of 19.56 kWh/100 km, but cannot adapt to new traffic patterns. A* path search algorithm [29] and Dijkstra algorithm [30] achieve an average energy consumption of 18.6 kWh/100 km by solving a static energy-cost graph in a single pass, yet they lack real-time adaptability to congestion or charging constraints. An Mixed-Integer Linear Programming (MILP) [31] formulation also achieves average energy consumption of 18.6 kWh/100 km, but at the expense of a solver runtime

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TABLE III

OPTIMAL PATHS AT VARIOUS CONGESTION WITHOUT USING BIPARTITE GRAPH IN FIG.2(A)

EV	Starting Position	Optimal Path with Battery SOC (kWh)	Energy Consumption (kWh)			Time (min.)		
			25% Congestion	50% Congestion	100% Congestion	25% Congestion	50% Congestion	100% Congestion
1	B	[B (8.50), E (6.50), G (4.50), CS2 (2.50), → recharge to 10.00, I (9.74), J (9.48), K (9.23), O (8.97), Q (8.71)]	6.78	7.29	8.54	74	83	96.6
2	H	[H (7.80), I (7.534), J (7.268), K (7.002), O (6.736), Q (6.47)]	1.25	1.33	1.42	42.5	51	61.2
3	D	[D (9.20), G (6.97), I (4.74), CS6 (2.50), → recharge to 10.00, J (9.84), K (9.68), O (9.52), Q (9.36)]	6.57	7.34	7.21	78.25	81.5	101.4
4	K	[K (3.00), O (2.75), CS7 (2.50), → recharge to 10.00, N (9.74), Q (9.48)]	0.97	1.03	1.14	62.35	71.7	83.9
5	C	[C (7.40), E (6.175), G (4.95), L (3.725), CS1 (2.50), → recharge to 10.00, L (9.435), Q (8.87)]	5.23	6.03	7.23	88	102	123

TABLE IV

OPTIMAL PATHS AT VARIOUS CONGESTION USING BIPARTITE GRAPH IN FIG.2(A)

EV	Starting Position	Optimal Path with SOC (kWh)	Energy Consumption (kWh)			Time (min.)		
			25% Congestion	50% Congestion	100% Congestion	25% Congestion	50% Congestion	100% Congestion
1	B	[B (7.00), E (4.75), G (2.50), CS1 (recharged to 10.00), L (9.58), Q (9.16)]	4.66	5.34	6.12	60	72	86.4
2	H	[H (9.00), I (8.73), J (8.47), K (8.20), O (7.94), Q (7.67)]	1.25	1.33	1.42	42.5	51	61.2
3	D	[D (5.00), G (2.50), CS1 (recharged to 10.00), L (8.69), Q (7.38)]	4.46	5.13	5.81	66.25	79.5	95.4
4	K	[K (3.30), O (2.50), CS4 (recharged to 10.00), Q (9.87)]	0.86	0.93	0.99	51.25	61.5	73.8
5	C	[C (6.00), E (5.13), G (4.25), L (3.38), CS3 (recharged to 10.00), L (9.10), Q (8.20)]	4.63	5.31	6.10	75	90	108

TABLE V

OPTIMAL PATHS AT VARIOUS CONGESTION USING BIPARTITE GRAPH IN FIG.2(B)

EV	Starting Position	Optimal Path with SOC (kWh)	Energy Consumption (kWh)			Time (min.)		
			25% Congestion	50% Congestion	100% Congestion	25% Congestion	50% Congestion	100% Congestion
1	16	[16 (8.00), 11 (5.25), CS3 (2.50), → recharge to 10.00, 9 (9.75), 25 (9.50), 12 (9.25)]	5.32	6.24	7.52	70	82	89.4
2	24	[24 (6.50), CS1 (2.50), → recharge to 10.00, 3 (9.91), 6 (9.82), 13 (9.73), 4 (9.64), 12 (9.55)]	2.55	4.43	2.42	32.5	62	71.3
3	6	[6 (7.50), CS6 (2.50), → recharge to 10.00, 18 (9.75), 4 (9.50), 12 (9.25)]	4.67	5.74	5.56	66.30	81.5	97.4
4	13	[13 (4.20), 7 (3.35), CS7 (2.50), → recharge to 10.00, 12 (9.77)]	1.72	1.93	2.99	52.34	62.5	74.8
5	0	[0 (8.50), 23 (5.50), CS2 (2.50), → recharge to 10.00, 12 (9.59)]	5.63	6.41	8.10	85	74	98

TABLE VI

COMPARISON OF ENERGY CONSUMPTION WITH RESPECT TO CONGESTION FACTOR (PROPOSED VS WORK [22])

Congestion	Average Velocity (km/hr)		Energy Consumption Rate (kWh/100km)		Energy Economy Difference (%)
	Proposed	work [22]	Proposed	work [22]	
100%	10	34.5	16.66±1.4	16.1±3.6	3.48
50%	25	35.1	15.64±1.6	15.7±2.5	0.63
25%	35	35.9	13.83±1.8	15.6±2.3	12.8

Note: The values following '±' represent the standard deviation.

TABLE VII

COMPARISON OF DIFFERENT EV ROUTING METHODS

Method	**Energy Consumption (kWh/100 km)**	**Convergence Time**	**Key Features**	**Limitations**
Clustering historical data [28]	19.56	precomputed	Leverages real driving patterns	Static; no adaptation to new conditions
A* path search algorithm [29]	18.6	single-pass	Globally optimal on static cost graph	Cannot adapt to dynamic traffic or charging constraints
Dijkstra algorithm [30]	18.6	single-pass	Guarantees shortest energy path under fixed costs	Static; no real-time congestion handling
MILP [31]	18.6	10 min (solver run)	Deterministic optimum with time-windows	High computational complexity; poor scalability
PSO-based Method [32]	21.6	30 min (100 iterations)	Effective multi-objective balance	Sensitive to parameters; may converge to local optima
Traditional RL (SARSA) [33]	36.8	2 h (500 epochs)	Learns from interaction; handles stochastic traffic	High sample complexity; large state-space issues
Deep RL [22]	15.7	5 h (6000 epochs)	Value approximation over continuous states	Heavy training cost; stability requires careful tuning
Proposed Q-Learning	15.64	3.5 h (10000 epochs)	Real-time adaptation; low hyperparameter count	Requires initial training; sensitive to reward design

of approximately ten minutes for moderate network sizes, limiting its scalability. While flexible in balancing multiple objectives, the Particle Swarm Optimization (PSO) [32] based heuristic approach consumes 21.6 kWh/100 km of energy and requires about 30 minutes for 100 iterations, often converging to suboptimal solutions when parameters are not finely tuned. Traditional RL method SARSA [33] reaches 36.8 kWh/100 km and requires roughly two hours (500 epochs) to converge, illustrating high sample complexity and large state-space challenges. A DRL model [28] reduces energy consumption to 15.7 kWh/100 km but demands around five hours of training (6,000 epochs) and careful hyperparameter tuning to ensure stability. Our proposed Q-learning approach improves it to 15.64 kWh/100 km, converging in approximately 3.5 hours (10,000 epochs). This represents a marginal but meaningful 0.36% reduction compared to DRL, achieved with fewer trainable parameters and lower computation overhead. Our Q-learning approach strikes an optimal balance, delivering

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the lowest energy consumption of all listed methods with a reasonable training time and limited complexity.

Table VIII presents KL divergence values, measuring the difference between SG-GAN-generated and real-world maps of road networks. Higher values indicate more distance, while lower values indicate closer similarity. For 29 nodes, Graph 1 shows the highest KL divergence (0.9824). In the 32-node case, divergence values are moderate. For 34 nodes, Graph 3 has the smallest distance (0.1231), closely similar to real data. In the present work, both model training and testing were done on the same road network extracted from OSM. Crucially, SG-GAN does not simply replicate this reference road network; rather, it learns the underlying connectivity patterns and synthesizes entirely new road networks. We then employ KLD to compare the edge-feature distributions of these SG-GAN-generated road networks against the real one to provide a rigorous test of whether SG-GAN has truly generalized the real road network's structural statistics rather than overfitting to specific routes. KLD value decreases during training, confirming that the generated layouts progressively converge to the real distribution. We adopt KLD to evaluate the similarity between the generated ones and the real road networks due to its sensitivity to global distributional discrepancies. Unlike Jensen–Shannon Divergence (JSD) [34], which is symmetric and bounded, and Graphlet/Motif Divergence [35] that captures only local subgraph mismatches, KLD asymmetrically penalizes distributional gaps, particularly when rare but important behaviors are underrepresented in the generated graphs. This makes it well-suited for evaluating edge-level distance-to-time ratio distributions in road networks. In Table IX, for a 29-node graph, KLD drops significantly from 0.9824 at 200 iterations to 0.6559 at 400 iterations, a 33.2% improvement. In contrast, JSD improves moderately from 0.5123 to 0.4685 (8.5%), and Graphlet Divergence from 0.2134 to 0.2082 (2.4%). This performance illustrates KLD's superior sensitivity to global improvements during training.

Table X shows a comparison where SG-GAN outperforms Graphormer on all four key metrics. In the case of KLD Score, SG-GAN achieves a 2.16 times lower value (0.67 vs. 1.45), indicating that its generated road networks more closely resemble real-world structures. Regarding computational efficiency, SG-GAN trains 2.6 times faster (8.5 vs. 22.1 hours), converges with 30% fewer samples, and its inference speed is 8 times faster (15ms vs. 120ms) compared to Graphormer. In connectivity rate, SG-GAN achieves 92% valid connected graphs, whereas Graphormer only reaches 76%. Regarding average Graph Edit Distance (GED), SG-GAN records 180 ± 20 edits whereas Graphormer records 450 ± 25 edits. This means SG-GAN requires 2.5 times fewer insertions/deletions to match the real graph compared to Graphormer, which demonstrates superior structural fidelity. This performance gap is primarily due to Graphormer's global attention over-smoothing, which fails to capture essential topological rules.

Finally, we conduct a performance comparison assessment. Numerical values are reported in the Table VI, with a 2.68% reduction in energy consumption compared to [22].

TABLE VIII

KL DISTANCE FOR DIFFERENT SG-GAN GENERATED GRAPH FOR DIFFERENT ITERATION VS REAL GRAPH

No. of Nodes	Graph 1 (200 Iterations)	Graph 2 (300 Iterations)	Graph 3 (400 Iterations)
29	0.9824	0.7265	0.6559
32	0.7664	0.7258	0.4973
34	0.2731	0.2116	0.1201

TABLE IX

DIVERGENCE MEASURES BETWEEN GENERATED AND REAL GRAPHS USING DIFFERENT METHODS (29 NODES)

Metric	200 Iterations	300 Iterations	400 Iterations
KLD [23]	0.9824	0.7265	0.6559
JSD [34]	0.5123	0.4887	0.4685
Graphlet/Motif Divergence [35]	0.2134	0.2105	0.2082

TABLE X

PERFORMANCE COMPARISON BETWEEN SG-GAN AND GRAPHORMER

Metric	SG-GAN	Graphormer[36]	Improvement on SG-GAN
Connectivity Rate	0.92 ± 0.03	0.76 ± 0.04	+21%
Diversity (KLD Score)	0.65 ± 0.08	1.45 ± 0.11	2.16× better
Graph Edit Distance (GED)	180 ± 20	450 ± 25	2.5× fewer edits
Training Time (hrs)	8.5 ± 0.6	22.1 ± 1.4	2.6× faster
Inference Time (sec)	15 ± 2	120 ± 8	8× faster

Note: The values following '±' represent the standard deviation.

A. Computational Overhead and Performance

The computational complexity of SG-GAN and Q-Learning for EV routing and CS allocation is summarized in Table XI.

- I: Number of training iterations
- $|V_e|$: Number of nodes in the road network
- b': Samples per node in SG-GAN's generator step
- z': Positive/negative samples per node in SG-GAN's discriminator step
- F_G: Computational cost per generator forward pass
- F_D: Computational cost per discriminator forward pass
- $|E_d|$: Number of edges in the road graph
- P_G, P_D: Number of parameters in the generator and discriminator
- q, z: Temporary storage for generated samples and discriminator inputs
- n: Number of EV agents
- $|A|$: Q-learning action space size
- s': High-traffic nodes in Q-learning (with bipartite matching cost $\mathcal{O}(s'^3)$)
- CS: Number of CS
- $|s|$: State-space size for each agent's Q-table

SG-GAN primarily focuses on graph-based learning, requiring Breadth-First Search (BFS) tree construction and iterative generator-discriminator training, while the proposed Q-Learning models multi-agent decision-making with high-dimensional state-action spaces. The training time for SG-GAN depends on the number of iterations and the size of the network. In each training iteration, every node in the network undergoes a series of generator and discriminator operations. Although some operations (like constructing complete BFS trees) could, in the worst case, incur high computational costs, these are typically executed only on demand or updated incrementally, and also SG-GAN Offline

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TABLE XI
COMPUTATIONAL COMPLEXITY

Algorithm	Time Complexity	Space Complexity
SG-GAN	$O\Big(I\lvert V_e\rvert(b'F_G + z'F_D)\Big) + O(\lvert V_e\rvert^2)$	$O\Big(\lvert V_e\rvert + \lvert E_d\rvert + P_G + P_D + \lvert V_e\rvert(q+z)\Big) + O(\lvert V_e\rvert^2)$
Multiagent Q-Learning	$O\Big(I(n\lvert A\rvert + \lvert V_e\rvert + n(s'^3 + CS))\Big)$	$O\Big(n\lvert s\rvert\lvert A\rvert + \lvert V_e\rvert\Big)$

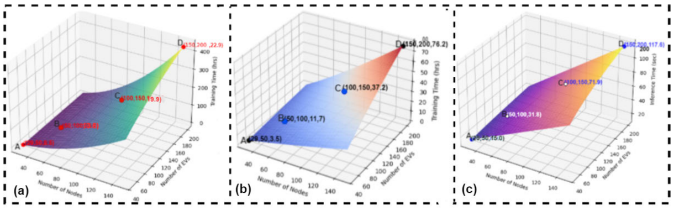

Fig. 6. Changes in Computational time by scaling road networks and No. of EVs (a) Training time for SG-GAN, (b) Training time for Q-Learning, (c) Inference time for graph generation using SG-GAN.

pre-training keeps the overall computational overhead within practical limits. For the present Q-Learning algorithm used in EV routing and CS allocation, the computational load increases with the number of EV agents, the size of their action space, and the number of nodes in the road network. Additionally, the algorithm handles extra tasks such as matching high-traffic nodes, which introduces further overhead. However, in parallel execution on our neural engine-based CPU, the algorithm ensures that the time required per iteration remains efficient, even as the problem size grows.

Fig.6(a) shows a 3D plot of SG-GAN training time with the following points: A (29 nodes, 50 EVs, 8.5 hrs), B (50 nodes, 100 EVs, 13.6 hrs), C (100 nodes, 150 EVs, 19.9 hrs), and D (150 nodes, 200 EVs, 22.9 hrs). The training time increases nearly quadratically with $|V_e|$ due to the dominant $O(|V_e|^2)$ term, while the impact of increasing EVs is less severe. Fig.6(b) presents the proposed Q-learning training time with points: A (29 nodes, 50 EVs, 3.5 hrs), B (50 nodes, 100 EVs, 11.7 hrs), C (100 nodes, 150 EVs, 37.2 hrs), and D (150 nodes, 200 EVs, 76.2 hrs). The effect of $|V_e|$, with larger networks is its increases the global state space, scanning cost, and connectivity management and even though the increase in EVs adds overhead, larger networks can accommodate this. Fig.6(c) depicts SG-GAN inference time for graph generation with points: A (29, 50, 8.5 sec), B (50, 100, 15 sec), C (100, 150, 71.9 sec), and D (150, 200, 117.6 sec). So in the present work, the scalability challenges for larger road networks can be addressed.

VII. CONCLUSION AND FUTURE WORKS

In conclusion, the present work proposes an SG-GAN-based road network development followed by a Q-learning-based decision-making strategy, thus effectively optimizing energy consumption in conjunction with enhancing EV routing and CS allocation performance. Representing the road network as a weighted graph, EVs navigate to fixed destinations while avoiding congestion and utilizing CS assigned via a bipartite graph approach. The solution achieves a 2.68% reduction in energy consumption and 0.97% KL distance in the generated road networks from the reference real one. Future work will integrate the proposed method and the work reported in [37] to leverage lightweight transformer-based collision detection and load estimation schemes, leading to improved routing reliability in scalable road networks under high-access loads.

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